Sergio Rodrigo Cárdenas Rivera

Full-Stack Software Engineer · Mobile · Data Science
Cochabamba, Bolivia · +591 77418588 · rodrigosergio93@gmail.com
LinkedIn: https://www.linkedin.com/in/srodrigo23/ · GitHub: https://github.com/srodrigo23 · Web: https://rodres.xyz/

SUMMARY

Full-stack Software Engineer with 4+ years of experience building web and mobile applications with TypeScript, React, Next.js, Nest.js, React Native and Python, plus experience in data engineering and machine learning. Led development teams as Tech Lead and Product Owner, shipping web and mobile products to production on AWS, and delivered solutions for international clients on the Salesforce platform. University programming instructor since 2014. Currently pursuing a Master's degree in Data Science and Intelligent Technologies.

EXPERIENCE

Full-Stack Software Engineer — Freelance May 2026 – Present · Cochabamba, Remote

Custom enterprise software development for local institutions.

- Gather and analyze functional and technical requirements with clients, turning them into web application specifications.
- Build and deploy full-stack web applications with Next.js, React and FastAPI, containerized with Docker.
- Streamlined clients' corporate information management, reducing operational workload and turnaround times.
- Train end users on the applications delivered.

Stack: Next.js, React, TypeScript, Redux, Zustand, Tailwind CSS, FastAPI, Docker

Tech Lead & Software Engineer — SHEMA YISRAEL S.R.L. May 2025 – May 2026 · Cochabamba, On-site

Head of the software department; Product Owner and technical lead of the development team.

- Led and coordinated the development team using agile methodologies, prioritizing the backlog as Product Owner.
- Designed and implemented web and mobile educational platforms for medical studies.
- Established quality standards and code review practices for the features delivered by the team.
- Developed, deployed and released web and mobile (React Native) applications to production on AWS (Amplify, S3, RDS).

Stack: Next.js, Nest.js, React, React Native, Redux, TypeScript, Tailwind CSS, PostgreSQL, Docker, AWS

Data Engineer — Alicorp (Peru) Aug 2024 – Nov 2024 · Remote

Customer analytics for the Bolivian operation of a Peruvian consumer goods company.

- Built ETL processes to extract, transform and load sales data from SQL Server.
- Cleaned and prepared customer purchase data from Bolivia's main economic corridor for unsupervised learning models.
- Segmented customers with clustering algorithms by purchase frequency, amount spent and payment time, as the basis for discount policies.
- Presented results through visualizations to support commercial decision-making.

Stack: Python, scikit-learn, matplotlib, SQL Server, PostgreSQL

Software Engineer — Oktana Corporation Dec 2021 – Feb 2024 · Remote

Software consultancy specialized in Salesforce, serving clients such as Sony PlayStation, Zola, Looker and Okta.

- Implemented the architecture of NLP conversational chatbots (Rasa) integrated with WhatsApp through the Meta API.
- Developed web applications with Python (Flask, SQLAlchemy) and React, containerized with Docker.
- Built applications with Next.js, TypeScript, Tailwind CSS, Prisma and PostgreSQL, deployed on Heroku and AWS S3.
- Developed Salesforce solution demos with Aura and Lightning Web Components for client presentations.
- Earned the Salesforce Developer I certification and worked with agile methodologies in distributed teams.

Stack: Python, Flask, SQLAlchemy, Rasa, React, Next.js, TypeScript, Prisma, PostgreSQL, Docker, Heroku, AWS S3, Salesforce (Apex, Aura, LWC)

SKILLS

- Languages: TypeScript, JavaScript, Python, Java, Kotlin, PHP, Apex, SQL, VBA
- Frontend: React, Next.js, Vue.js, Redux, Zustand, Tailwind CSS, HTML, CSS
- Backend: Nest.js, Node.js, Express, FastAPI, Flask, SQLAlchemy, Prisma
- Mobile: React Native, Android (Java, Kotlin)
- Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Firebase
- Cloud & DevOps: AWS (Amplify, Lambda, RDS, S3), Docker, Heroku, Git

- Data & ML: scikit-learn, Pandas, NumPy, matplotlib, Power BI, Rasa (NLP)
- Salesforce: Apex, Lightning Web Components, Aura Components, SOQL, SOSL
- Practices: Agile methodologies, code review, requirements gathering

EDUCATION

- Master's Degree in Data Science and Intelligent Technologies — Universidad Técnica de Oruro | In progress
- Postgraduate Diploma in Competency-Based Higher Education — Universidad Mayor de San Simón (UMSS) | Sep 2021. Final project: course syllabus for Programming Fundamentals and Data Structures.
- Bachelor's Degree in Informatics Engineering — UMSS | Jan 2018 – Dec 2021. Capstone: Video Surveillance Smart System, a distributed surveillance system using neural networks to detect people and alert to hazardous situations at home.
- Bachelor's Degree in Systems Engineering — UMSS | Jan 2011 – Dec 2017. Capstone: Media Box Player, an Android app (Java, MVP architecture, ExoPlayer, Dropbox API) for streaming multimedia content from the cloud.

TEACHING (UMSS — FACULTY OF SCIENCE AND TECHNOLOGY, COCHABAMBA)

- University Lecturer, intensive term courses | Aug 2023 – Jan 2025
  - Information Systems 2 (Jan 2025, 80 students): information systems implementation, teamwork and agile methodologies.
  - Computing 1 (Aug 2023 and Jan 2024, 110 students per course): programming fundamentals with VBA.
- Teaching Assistant | Feb 2014 – Dec 2018
  - Computing 1: 8 semesters (2014–2017) and 4 intensive term courses (2015–2018), 100–150 students per course; basic programming with VBA.
  - Introduction to Programming: 2 semesters (2015), 80 students; programming with Java.
- Instructor | Feb 2014 – Dec 2018
  - 8 VBA-in-Excel programming lab courses, 50–80 students each.
  - 2 office productivity courses (Word, Excel), 20–30 participants, for the Seventh Division of the Bolivian Army (Dec 2014).

ADDITIONAL EXPERIENCE (DEPARTMENTAL ELECTORAL TRIBUNAL)

- Election Results Transmitter | Mar 2026 — Transmitted polling-station results in real time through the SIREPRE mobile app during the 2026 Subnational Elections.
- Electoral Records Digitizer | Dec 2024 — Digitized tally sheets for the quick count of the 2024 Judicial Branch and Constitutional Court Elections.

CERTIFICATIONS

- Frontend Developer (React) — HackerRank | Feb 2026 | https://www.hackerrank.com/certificates/ec0617594afe
- React (Basic) — HackerRank | Feb 2026 | https://www.hackerrank.com/certificates/af68fd41c99e
- Power BI Basic and Intermediate — UMSS | Jun 2023
- Crash Course on Python — Coursera (Google) | Jan 2023 | https://coursera.org/share/93b9d9abe585cad1e39e8ca427032db2
- Trailhead Ranger — Salesforce Trailhead | Apr 2022 | https://www.salesforce.com/trailblazer/srodrigo23
- Salesforce Developer I — Salesforce | Mar 2022 | https://www.salesforce.com/trailblazer/srodrigo23
- Python (Basic) — HackerRank | Jan 2022 | https://www.hackerrank.com/certificates/bf595008a8ce
- Java (Basic) — HackerRank | Jul 2020 | https://www.hackerrank.com/certificates/0c54c3e28f00
- Problem Solving (Basic) — HackerRank | Jul 2020 | https://www.hackerrank.com/certificates/2f7a6e9672d8

COMMUNITY & EVENTS

- Mentor — Tech4Future Hackathon 2026 — UMSS · Bolivian Hub of AI, Cochabamba | Feb 2026
- Mentor — Google Hackathon "Build with AI" — Universidad Católica Boliviana, Santa Cruz | Jul 2025
- Seminars attended (2011–2013, UMSS and others): Information Security Conferences (2nd and 3rd editions), Law 164 on Telecommunications/ICT and Digital Signature, ICT-based content creation in Bolivia, Túpac Katari communications satellite, Microsoft Exchange Server 2010 (C.O.G.N.O.S.), Java Programming (TEKHNE S.R.L.).

LANGUAGES

Spanish (native) · English (upper-intermediate, B2)